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Faculty of Mechanical and Industrial Engineering

Fluid Mechanics (Meng 3171)

Chapter FOUR

Differential Relations for a Fluid Flow

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Outline

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- The Acceleration Field of a Fluid
- The Differential Equation of Mass Conservation
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- The Stream Function
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- Frictionless Irrotational Flows



Introduction

- In analyzing fluid motion, we might take one of two paths:
 - ❖ seeking an estimate of gross effects (**integral relation**) over a finite region or control volume
 - ❖ seeking the point-by-point details of a flow pattern by analyzing an infinitesimal region of the flow (**Differential relation**).



The Acceleration Field of a Fluid

- The cartesian vector form of a velocity field that varies in space and time is;
- $V_{(r,t)} = iu(x, y, z, t) + jv(x, y, z, t) + kw(x, y, z, t)$
- This is the most important variable in fluid mechanics: Knowledge of the velocity vector field is nearly equivalent to solving a fluid flow problem.
- Our coordinates are fixed in space, and we observe the fluid as it passes by—as if we had scribed a set of coordinate lines on a glass window in a wind tunnel. This is **the eulerian fram of reference**.



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- To calculate the acceleration vector field we compute the total derivative of the velocity field;
- $a = \frac{dV}{dt} = i \frac{du}{dt} + j \frac{dv}{dt} + k \frac{dw}{dt}$

we use the chain rule to obtain each scalar time derivative. That is;

$$\frac{du(x, y, z, t)}{dt} = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt} + \frac{\partial u}{\partial z} \frac{dz}{dt}$$
$$dy/dt = v,$$



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$$\begin{aligned}a_x &= \frac{du}{dt} = \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = \frac{\partial u}{\partial t} + (\mathbf{V} \cdot \nabla)u \\a_y &= \frac{dv}{dt} = \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} = \frac{\partial v}{\partial t} + (\mathbf{V} \cdot \nabla)v \\a_z &= \frac{dw}{dt} = \frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} = \frac{\partial w}{\partial t} + (\mathbf{V} \cdot \nabla)w\end{aligned}$$

Summing these into a vector, we obtain the total acceleration:

$$\mathbf{a} = \frac{d\mathbf{V}}{dt} = \underbrace{\frac{\partial \mathbf{V}}{\partial t}}_{\text{Local}} + \underbrace{\left(u \frac{\partial \mathbf{V}}{\partial x} + v \frac{\partial \mathbf{V}}{\partial y} + w \frac{\partial \mathbf{V}}{\partial z} \right)}_{\text{Convective}} = \frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V} \cdot \nabla)\mathbf{V}$$



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- The **local acceleration** vanishes if the flow is steady—that is, independent of time.
- The **convective acceleration** arises when the particle moves through regions of spatially varying velocity, as in a nozzle or diffuser.

$$u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y} + w \frac{\partial}{\partial z} = \mathbf{V} \cdot \nabla$$

$$\text{where} \quad \nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$$



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Example: Given the eulerian velocity vector field $V = 3ti + xzj + ty^2k$, find the total acceleration of a particle.

- Solution step 1: work out the local acceleration $\partial V / \partial t$

$$\frac{\partial \mathbf{V}}{\partial t} = \mathbf{i} \frac{\partial u}{\partial t} + \mathbf{j} \frac{\partial v}{\partial t} + \mathbf{k} \frac{\partial w}{\partial t} = \mathbf{i} \frac{\partial}{\partial t} (3t) + \mathbf{j} \frac{\partial}{\partial t} (xz) + \mathbf{k} \frac{\partial}{\partial t} (ty^2) = 3\mathbf{i} + 0\mathbf{j} + y^2\mathbf{k}$$

- Solution step 2: convective acceleration

$$u \frac{\partial \mathbf{V}}{\partial x} = (3t) \frac{\partial}{\partial x} (3t\mathbf{i} + xz\mathbf{j} + ty^2\mathbf{k}) = (3t)(0\mathbf{i} + z\mathbf{j} + 0\mathbf{k}) = 3tz\mathbf{j}$$

$$v \frac{\partial \mathbf{V}}{\partial y} = (xz) \frac{\partial}{\partial y} (3t\mathbf{i} + xz\mathbf{j} + ty^2\mathbf{k}) = (xz)(0\mathbf{i} + 0\mathbf{j} + 2ty\mathbf{k}) = 2txyz\mathbf{k}$$

$$w \frac{\partial \mathbf{V}}{\partial z} = (ty^2) \frac{\partial}{\partial z} (3t\mathbf{i} + xz\mathbf{j} + ty^2\mathbf{k}) = (ty^2)(0\mathbf{i} + x\mathbf{j} + 0\mathbf{k}) = txy^2\mathbf{j}$$



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- Solution step 3: combine all forms

$$\begin{aligned}\frac{dV}{dt} &= \frac{\partial V}{\partial t} + u \frac{\partial V}{\partial x} + v \frac{\partial V}{\partial y} + w \frac{\partial V}{\partial z} = (3\mathbf{i} + y^2\mathbf{k}) + 3tz\mathbf{j} + 2txyz\mathbf{k} + txy^2\mathbf{j} \\ &= 3\mathbf{i} + (3tz + txy^2)\mathbf{j} + (y^2 + 2txyz)\mathbf{k}\end{aligned}$$

Comments: Assuming that V is valid everywhere as given, this total acceleration vector dV/dt applies to all positions and times within the flow field.



The Differential Equation of Mass Conservation

- Conservation of mass, often called the continuity relation, states that the fluid mass cannot change. All the basic differential equations can be derived by considering either **an elemental control volume** or **an elemental system**.
- We choose an infinitesimal fixed control volume (dx, dy, dz), from chapter 3

$$\int_{CV} \frac{\partial \rho}{\partial t} d\mathcal{V} + \sum_i (\rho_i A_i V_i)_{out} - \sum_i (\rho_i A_i V_i)_{in} = 0$$

- The element is so small that the volume integral simply reduces to a differential term:

$$\int_{CV} \frac{\partial \rho}{\partial t} d\mathcal{V} \approx \frac{\partial \rho}{\partial t} dx dy dz$$

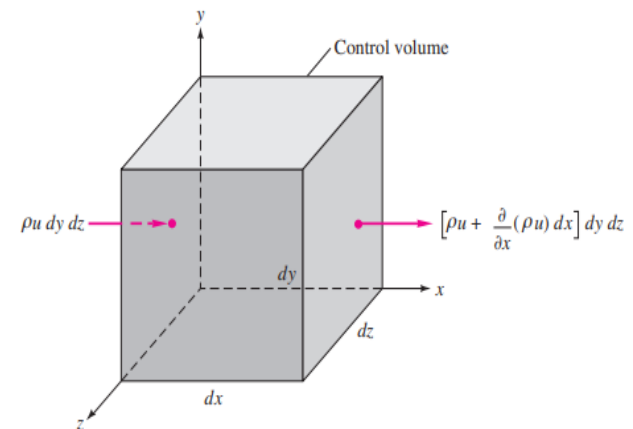


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- The mass flow terms occur on all six faces, three inlets and three outlets.

Face	Inlet mass flw	Outlet mass flw
x	$\rho u \, dy \, dz$	$\left[\rho u + \frac{\partial}{\partial x} (\rho u) \, dx \right] dy \, dz$
y	$\rho v \, dx \, dz$	$\left[\rho v + \frac{\partial}{\partial y} (\rho v) \, dy \right] dx \, dz$
z	$\rho w \, dx \, dy$	$\left[\rho w + \frac{\partial}{\partial z} (\rho w) \, dz \right] dx \, dy$

- Introduce these terms into above eqn.
- and we have;





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- Introduce these terms into above eqn. and we have;

$$\frac{\partial \rho}{\partial t} dx dy dz + \frac{\partial}{\partial x} (\rho u) dx dy dz + \frac{\partial}{\partial y} (\rho v) dx dy dz + \frac{\partial}{\partial z} (\rho w) dx dy dz = 0$$

The element volume cancels out of all terms, leaving a partial differential equation involving the derivatives of density and velocity:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x} (\rho u) + \frac{\partial}{\partial y} (\rho v) + \frac{\partial}{\partial z} (\rho w) = 0$$

Continuum concept all fluid properties are considered to be uniformly varying functions of time and position such as (x, y, z, t).



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- It is often called the equation of continuity because it requires no assumptions except that the density and velocity are **continuum functions**. That is, the flow may be either **steady or unsteady, viscous or frictionless, compressible or incompressible**.

- The divergence of the vector $\rho \mathbf{V}$;

$$\frac{\partial}{\partial x} (\rho u) + \frac{\partial}{\partial y} (\rho v) + \frac{\partial}{\partial z} (\rho w) \equiv \nabla \cdot (\rho \mathbf{V})$$

- General Cartesian compact form of the continuity relation is;

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0$$

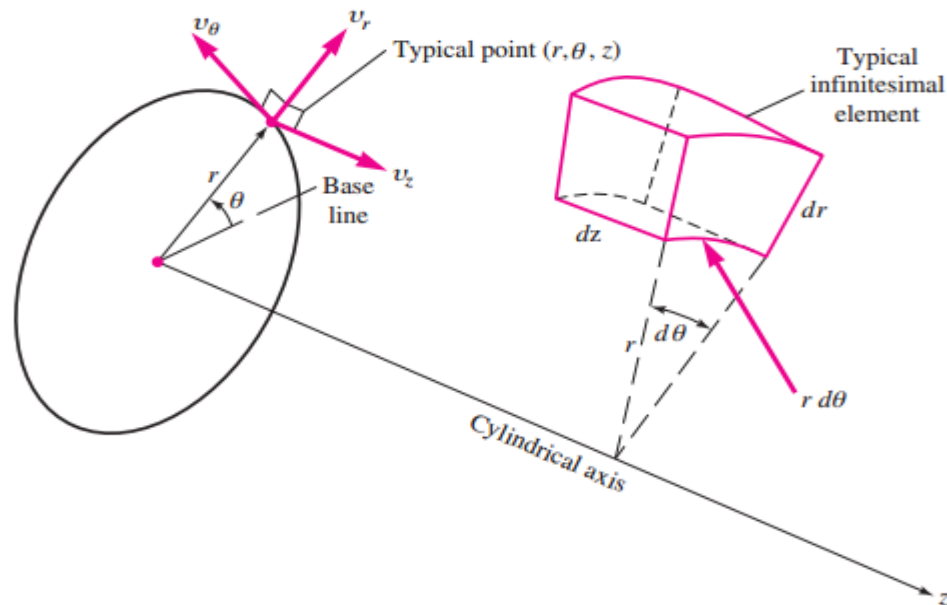
- The general continuity equation in cylindrical polar coordinates is;



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- The general continuity equation in cylindrical polar coordinates is;

$$\frac{\partial \rho}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (r \rho v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (\rho v_\theta) + \frac{\partial}{\partial z} (\rho v_z) = 0$$





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Cartesian:
$$\frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0$$

Cylindrical:
$$\frac{1}{r} \frac{\partial}{\partial r}(r \rho v_r) + \frac{1}{r} \frac{\partial}{\partial \theta}(\rho v_\theta) + \frac{\partial}{\partial z}(\rho v_z) = 0$$

Incompressible; special case that affords great simplification is incompressible flow, where the density changes are negligible.

$$\partial \rho / \partial t \approx 0$$

- The result is valid for steady or unsteady incompressible flow.

Cartesian:
$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

Cylindrical:
$$\frac{1}{r} \frac{\partial}{\partial r}(r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta}(v_\theta) + \frac{\partial}{\partial z}(v_z) = 0$$



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Example: An incompressible velocity field is given by

$u=a(x^2-y^2)$, v unknown and $w=b$. where a and b are constants. What must the form of the velocity component v be?

Solution:

Example: The velocity components for a certain incompressible, steady-flow field are $u=x^2+y^2+z^2$, $v=xy+yz+z$ and w =unkown. Determine the form of the z component, w , required to satisfy the continuity equation.



The Differential Equation of Linear Momentum

- The appropriate form of the linear momentum relation is;

$$\sum \mathbf{F} = \frac{\partial}{\partial t} \left(\int_{CV} \mathbf{V} \rho d^3V \right) + \sum (\dot{m}_i \mathbf{V}_i)_{out} - \sum (\dot{m}_i \mathbf{V}_i)_{in}$$

- Again the element is so small that the volume integral simply reduces to a derivative term:

$$\frac{\partial}{\partial t} (\mathbf{V} \rho d^3V) \approx \frac{\partial}{\partial t} (\rho \mathbf{V}) dx dy dz$$

- The momentum fluxes occur on all six faces, three inlets and three outlets.

Faces	Inlet momentum flx	Outlet momentum flx
x	$\rho u \mathbf{V} dy dz$	$\left[\rho u \mathbf{V} + \frac{\partial}{\partial x} (\rho u \mathbf{V}) dx \right] dy dz$
y	$\rho v \mathbf{V} dx dz$	$\left[\rho v \mathbf{V} + \frac{\partial}{\partial y} (\rho v \mathbf{V}) dy \right] dx dz$
z	$\rho w \mathbf{V} dx dy$	$\left[\rho w \mathbf{V} + \frac{\partial}{\partial z} (\rho w \mathbf{V}) dz \right] dx dy$



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- The relation becomes;

$$\sum \mathbf{F} = dx dy dz \left[\frac{\partial}{\partial t} (\rho \mathbf{V}) + \frac{\partial}{\partial x} (\rho u \mathbf{V}) + \frac{\partial}{\partial y} (\rho v \mathbf{V}) + \frac{\partial}{\partial z} (\rho w \mathbf{V}) \right]$$

- Note that this is a vector relation. A simplification occurs if we split up the term in brackets as follows:

$$\begin{aligned} & \frac{\partial}{\partial t} (\rho \mathbf{V}) + \frac{\partial}{\partial x} (\rho u \mathbf{V}) + \frac{\partial}{\partial y} (\rho v \mathbf{V}) + \frac{\partial}{\partial z} (\rho w \mathbf{V}) \quad \text{Acceleration} \\ &= \mathbf{V} \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) \right] + \rho \left(\frac{\partial \mathbf{V}}{\partial t} + u \frac{\partial \mathbf{V}}{\partial x} + v \frac{\partial \mathbf{V}}{\partial y} + w \frac{\partial \mathbf{V}}{\partial z} \right) \end{aligned}$$



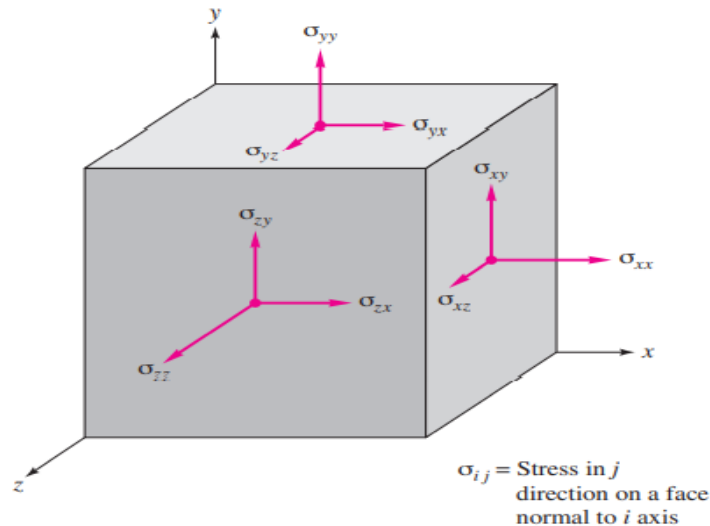
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- Thus we have now reduced to; $\sum \mathbf{F} = \rho \frac{d\mathbf{V}}{dt} dx dy dz$

$$\sum \mathbf{F} = \boxed{\text{Body force}} + \boxed{\text{Surface Force}}$$

$$d\mathbf{F}_{\text{grav}} = \rho \mathbf{g} dx dy dz$$

$$\sigma_{ij} = \begin{vmatrix} -p + \tau_{xx} & \tau_{yx} & \tau_{zx} \\ \tau_{xy} & -p + \tau_{yy} & \tau_{zy} \\ \tau_{xz} & \tau_{yz} & -p + \tau_{zz} \end{vmatrix}$$





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- The net surface force in the x direction is given by;

$$dF_{x,\text{surf}} = \left[\frac{\partial}{\partial x} (\sigma_{xx}) + \frac{\partial}{\partial y} (\sigma_{yx}) + \frac{\partial}{\partial z} (\sigma_{zx}) \right] dx dy dz$$

- Notice that the stress terms are taken from the top row of the array. Splitting this row into pressure plus viscous stresses, we can rewrite

$$\frac{dF_x}{dV} = -\frac{\partial p}{\partial x} + \frac{\partial}{\partial x} (\tau_{xx}) + \frac{\partial}{\partial y} (\tau_{yx}) + \frac{\partial}{\partial z} (\tau_{zx})$$

$$\frac{dF_y}{dV} = -\frac{\partial p}{\partial y} + \frac{\partial}{\partial x} (\tau_{xy}) + \frac{\partial}{\partial y} (\tau_{yy}) + \frac{\partial}{\partial z} (\tau_{zy})$$

$$\frac{dF_z}{dV} = -\frac{\partial p}{\partial z} + \frac{\partial}{\partial x} (\tau_{xz}) + \frac{\partial}{\partial y} (\tau_{yz}) + \frac{\partial}{\partial z} (\tau_{zz})$$



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- We can write the equation in short form as;

$$\left(\frac{d\mathbf{F}}{d\mathcal{V}}\right)_{\text{surf}} = -\nabla p + \left(\frac{d\mathbf{F}}{d\mathcal{V}}\right)_{\text{viscous}}$$

- where the viscous force has a total of nine terms:

$$\left(\frac{d\mathbf{F}}{d\mathcal{V}}\right)_{\text{viscous}} = \nabla \cdot \boldsymbol{\tau}_{ij}$$

$$\boldsymbol{\tau}_{ij} = \begin{bmatrix} \tau_{xx} & \tau_{yx} & \tau_{zx} \\ \tau_{xy} & \tau_{yy} & \tau_{zy} \\ \tau_{xz} & \tau_{yz} & \tau_{zz} \end{bmatrix}$$

$$\begin{aligned} \left(\frac{d\mathbf{F}}{d\mathcal{V}}\right)_{\text{viscous}} &= \mathbf{i} \left(\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \right) \\ &+ \mathbf{j} \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} \right) \\ &+ \mathbf{k} \left(\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \tau_{zz}}{\partial z} \right) \end{aligned}$$



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- We have the basic differential momentum equation for an infinitesimal element:

$$\rho \mathbf{g} - \nabla p + \nabla \cdot \boldsymbol{\tau}_{ij} = \rho \frac{d\mathbf{V}}{dt}$$

$$\frac{d\mathbf{V}}{dt} = \frac{\partial \mathbf{V}}{\partial t} + u \frac{\partial \mathbf{V}}{\partial x} + v \frac{\partial \mathbf{V}}{\partial y} + w \frac{\partial \mathbf{V}}{\partial z}$$

- Gravity force per unit volume + pressure force per unit volume + viscous force per unit volume = density* acceleration



Inviscid Flow: Euler's Equation

- The simplest assumption is frictionless flow $\tau_{ij} = 0$;

$$\rho \mathbf{g} - \nabla p = \rho \frac{d\mathbf{V}}{dt}$$

- This is **Euler's equation** for inviscid flow.

Newtonian Fluid: Navier-Stokes Equations

- For a Newtonian fluid, the viscous stresses are proportional to the element strain rates and the coefficient of viscosity.

$$\begin{aligned}\tau_{xx} &= 2\mu \frac{\partial u}{\partial x} & \tau_{yy} &= 2\mu \frac{\partial v}{\partial y} & \tau_{zz} &= 2\mu \frac{\partial w}{\partial z} \\ \tau_{xy} &= \tau_{yx} = \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) & \tau_{xz} &= \tau_{zx} = \mu \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right) \\ \tau_{yz} &= \tau_{zy} = \mu \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right)\end{aligned}$$



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- The differential momentum equation for a Newtonian fluid with constant density and viscosity:

$$\begin{aligned}\rho g_x - \frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) &= \rho \frac{du}{dt} \\ \rho g_y - \frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) &= \rho \frac{dv}{dt} \\ \rho g_z - \frac{\partial p}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) &= \rho \frac{dw}{dt}\end{aligned}$$

- These are the incompressible flow **Navier-Stokes** equations.
- They are second-order nonlinear partial differential equations and are quite formidable, but solutions have been found to a variety of interesting viscous flow problems.



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- **Example:** Take the velocity field $u=a(x^2-y^2)$, $v=-2axy$ and $w=0$, determine under what conditions it is a solution to the Navier-Stokes momentum equations (4.38). Assuming that these conditions are met, determine the resulting pressure distribution. Where $g_x=g_y=0$ and $g_z=-g$.
- **Solution:**



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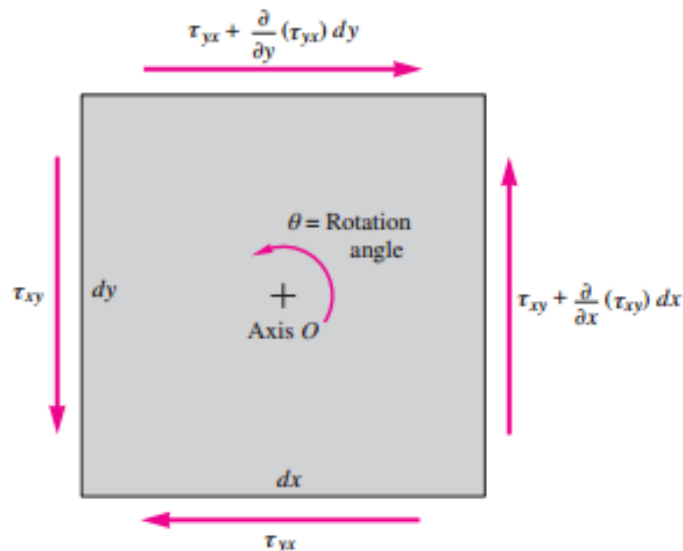
- **Example:** Consider the steady, two-dimensional, incompressible velocity field , namely, $V = (u, v) = (ax + b)i + (-ay + cx)j$. Calculate the pressure as a function of x and y .
- Solution:



The Differential Equation of Angular Momentum

- There is no differential angular momentum equation. Application of the integral theorem to a differential element gives the result, well known to students of stress analysis or strength of materials, that the shear stresses are symmetric:

$$\tau_{ij} = \tau_{ji}$$





The Differential Equation of Energy

The appropriate integral relation for the fixed control

Volume is;

$$\dot{Q} - \dot{W}_s - \dot{W}_v = \frac{\partial}{\partial t} \left(\int_{CV} e \rho dV \right) + \int_{CS} \left(e + \frac{p}{\rho} \right) \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

Since there is no shaft work W_s

$$\dot{Q} - \dot{W}_v = \left[\frac{\partial}{\partial t}(\rho e) + \frac{\partial}{\partial x}(\rho u \zeta) + \frac{\partial}{\partial y}(\rho v \zeta) + \frac{\partial}{\partial z}(\rho w \zeta) \right] dx dy dz$$

When we use the continuity equation by analogy

where $\zeta = e + p/\rho$.

$$\dot{Q} - \dot{W}_v = \left(\rho \frac{de}{dt} + \mathbf{V} \cdot \nabla p + p \nabla \cdot \mathbf{V} \right) dx dy dz$$



Boundary Conditions for the Basic Equations

- There are three basic differential equations of fluid motion;

Continuity:
$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0$$

Momentum:
$$\rho \frac{d\mathbf{V}}{dt} = \rho \mathbf{g} - \nabla p + \nabla \cdot \boldsymbol{\tau}_{ij}$$

Energy:
$$\rho \frac{d\hat{u}}{dt} + p(\nabla \cdot \mathbf{V}) = \nabla \cdot (k \nabla T) + \Phi$$

$$\begin{aligned} \Phi = \mu & \left[2 \left(\frac{\partial u}{\partial x} \right)^2 + 2 \left(\frac{\partial v}{\partial y} \right)^2 + 2 \left(\frac{\partial w}{\partial z} \right)^2 + \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right)^2 \right. \\ & \left. + \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right)^2 + \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right)^2 \right] \end{aligned}$$



Boundary Conditions for the Basic Equations

- There are five variables and we need two relations to solve the system of equations, that is state relation for thermodynamic properties;

$$\rho = \rho(p, T) \quad \hat{u} = \hat{u}(p, T)$$

- For example, for a perfect gas with constant specific heats, we complete the system with

$$\rho = \frac{p}{RT} \quad \hat{u} = \int c_v dT \approx c_v T + \text{const}$$

- What are the proper boundary conditions?

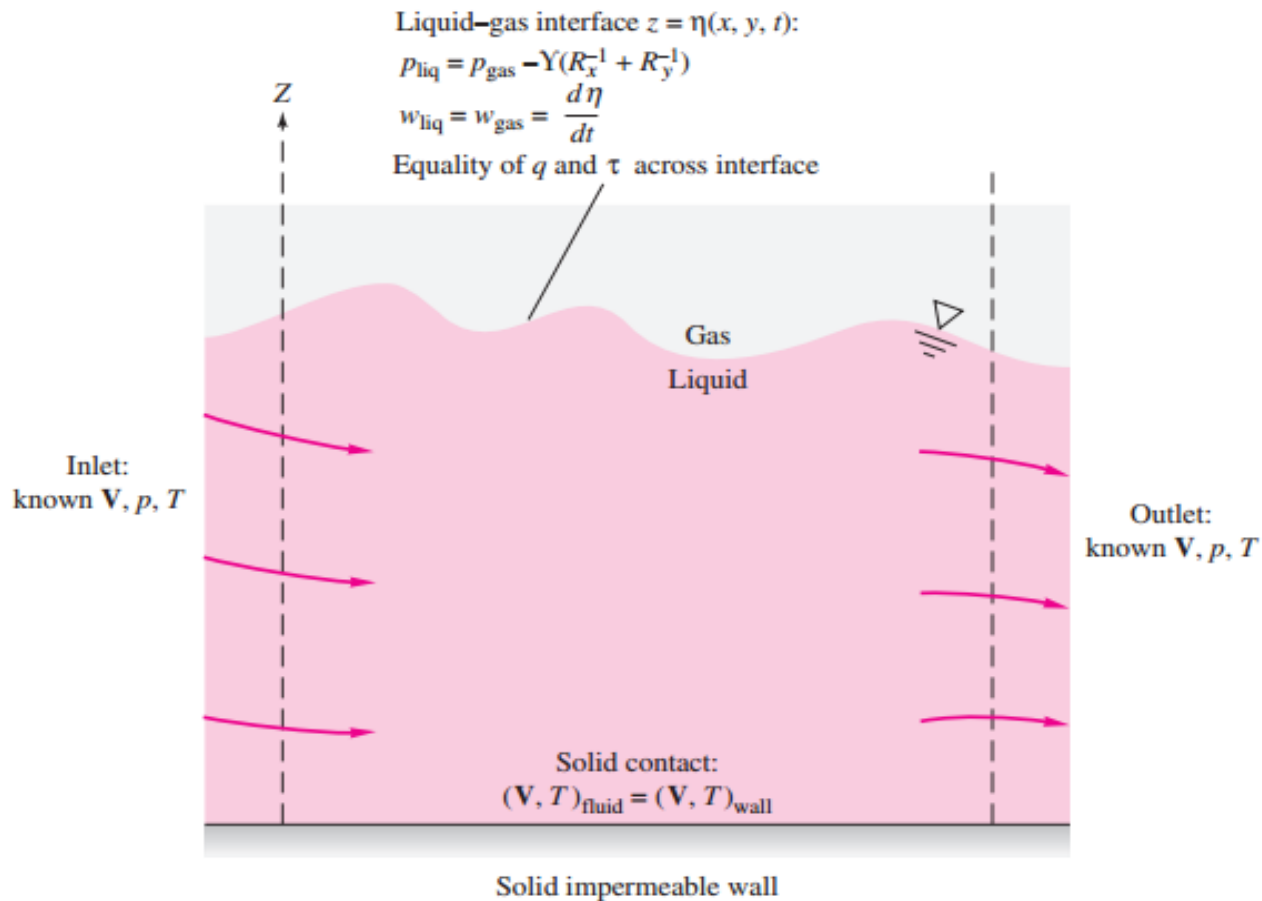


Boundary Conditions for the Basic Equations

- For unsteady state the initial conditions should be given. At $t=0$, $\rho, v, p = \text{known } f(x, y, z)$
- Solid wall $V_{\text{fluid}} = V_{\text{wall}}, T_{\text{fluid}} = T_{\text{wall}}$
- Inlet or outlet known V, p
- At the interface between liquid and gas



Boundary Conditions for the Basic Equations





The Stream Function

- The **stream function ψ** is a clever device that allows us to satisfy **the continuity equation** and then **solve the momentum equation** directly for the single variable ψ . Lines of constant ψ are stream lines of the flow.
- The stream function idea works only if the continuity equation can be reduced to two terms. The most common application is **incompressible flow in the xy plane**.

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

This equation is satisfied identically if a function $\psi(x, y)$ is defined as;

$$\frac{\partial}{\partial x} \left(\frac{\partial \psi}{\partial y} \right) + \frac{\partial}{\partial y} \left(-\frac{\partial \psi}{\partial x} \right) \equiv 0$$



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$$u = \frac{\partial \psi}{\partial y} \quad v = -\frac{\partial \psi}{\partial x}$$

$$\mathbf{V} = \mathbf{i} \frac{\partial \psi}{\partial y} - \mathbf{j} \frac{\partial \psi}{\partial x}$$

The definition of a streamline in two-dimensional flow is;

$$\frac{dx}{u} = \frac{dy}{v}$$
$$u \, dy - v \, dx = 0 \quad \text{streamline}$$

$$\frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy = 0 = d\psi$$

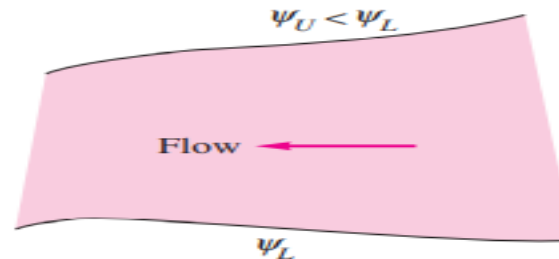
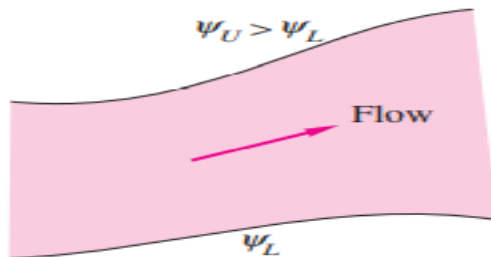
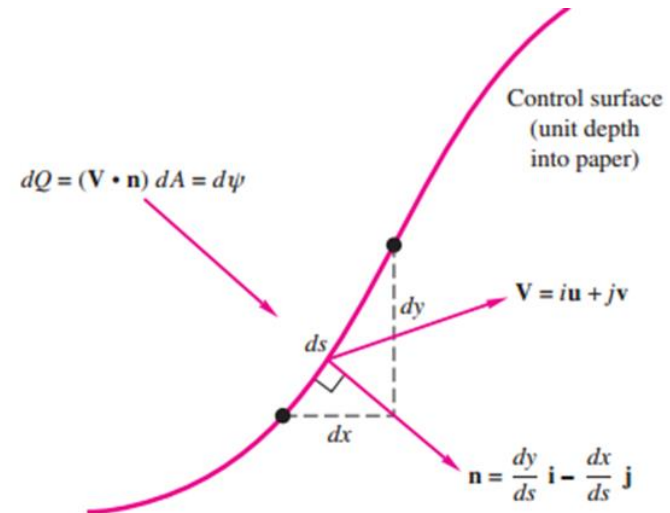
- Thus the change in ψ is zero along a streamline, or $\psi = \text{constant}$ along a streamline.
- The volume flow between any two streamlines in the flow field is equal to the change in stream function between those streamlines:



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$$dQ = (\mathbf{V} \cdot \mathbf{n}) dA = \left(\mathbf{i} \frac{\partial \psi}{\partial y} - \mathbf{j} \frac{\partial \psi}{\partial x} \right) \cdot \left(\mathbf{i} \frac{dy}{ds} - \mathbf{j} \frac{dx}{ds} \right) ds(1)$$

$$= \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy = d\psi$$



(a) flow to the right if ψ_U is greater; (b) flow to the left if ψ_L is greater.

$$Q_{1 \rightarrow 2} = \int_1^2 (\mathbf{V} \cdot \mathbf{n}) dA = \int_1^2 d\psi = \psi_2 - \psi_1$$



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Steady Plane Compressible Flow

- The equation of continuity becomes $\frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) = 0$
- Therefore a compressible flow stream function can be defined such that $\rho u = \frac{\partial \psi}{\partial y} \quad \rho v = -\frac{\partial \psi}{\partial x}$
- Again lines of constant ψ are streamlines of the flow, but the change in ψ is now equal to the mass flow, not the volume flow:

$$d\dot{m} = \rho(\mathbf{V} \cdot \mathbf{n}) dA = d\psi$$

$$\dot{m}_{1 \rightarrow 2} = \int_1^2 \rho(\mathbf{V} \cdot \mathbf{n}) dA = \psi_2 - \psi_1$$



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Incompressible Plane Flow in Polar Coordinates; Suppose that the important coordinates are r and θ , with $v_z = 0$, and that the density is constant.

$$\frac{\partial}{\partial r} \left(\frac{\partial \psi}{\partial \theta} \right) + \frac{\partial}{\partial \theta} \left(-\frac{\partial \psi}{\partial r} \right) = 0$$

$$\frac{1}{r} \frac{\partial}{\partial r} (r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (v_\theta) = 0$$

$$v_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} \quad v_\theta = -\frac{\partial \psi}{\partial r}$$



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- **Incompressible Axisymmetric Flow**; suppose that the flow is three-dimensional (v_r , v_z) but with no circumferential variations;

$$v_\theta = \frac{\partial}{\partial \theta} = 0$$

$$\frac{1}{r} \frac{\partial}{\partial r}(r v_r) + \frac{\partial}{\partial z}(v_z) = 0$$

$$\frac{\partial}{\partial r}(r v_r) + \frac{\partial}{\partial z}(r v_z) = 0$$

$$v_r = -\frac{1}{r} \frac{\partial \psi}{\partial z} \quad v_z = \frac{1}{r} \frac{\partial \psi}{\partial r}$$

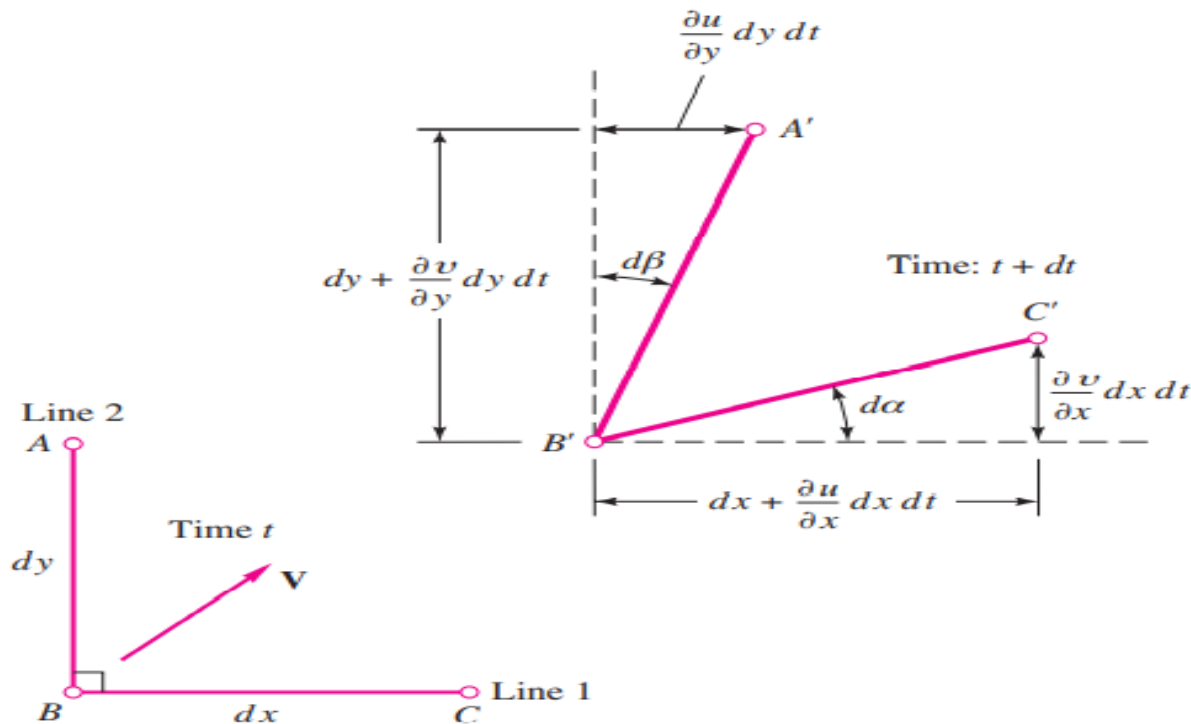
$$\frac{\partial}{\partial r} \left(-\frac{\partial \psi}{\partial z} \right) + \frac{\partial}{\partial z} \left(\frac{\partial \psi}{\partial r} \right) = 0$$

$$Q_{1 \rightarrow 2} = 2\pi(\psi_2 - \psi_1)$$



Vorticity and Irrotationality

The assumption of zero fluid angular velocity is called irrotationality. Here we show that **angular velocity** is associated with **the curl of the local velocity vector**.





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We define the angular velocity ω_z about the z axis as the average rate of counterclockwise turning of the two lines:

$$\omega_z = \frac{1}{2} \left(\frac{d\alpha}{dt} - \frac{d\beta}{dt} \right)$$

$$d\alpha = \lim_{dt \rightarrow 0} \left[\tan^{-1} \frac{(\partial v / \partial x) dx dt}{dx + (\partial u / \partial x) dx dt} \right] = \frac{\partial v}{\partial x} dt$$

$$d\beta = \lim_{dt \rightarrow 0} \left[\tan^{-1} \frac{(\partial u / \partial y) dy dt}{dy + (\partial v / \partial y) dy dt} \right] = \frac{\partial u}{\partial y} dt$$

$$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \quad \omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \quad \omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right)$$



...Cont'd

The vector $\omega = i\omega_x + j\omega_y + k\omega_z$ is thus one-half the curl of the velocity vector.

$$\omega = \frac{1}{2}(\text{curl } \mathbf{V}) = \frac{1}{2} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix}$$

- The vector twice larger is called vorticity. $\zeta = 2\omega = \text{curl } \mathbf{V}$
- Many flows have negligible or zero vorticity and are called irrotational. Such flows can be incompressible or compressible, steady or unsteady.

$$\text{curl } \mathbf{V} = 0$$



Frictionless Irrotational Flows

- When a flow is both frictionless and irrotational, pleasant things happen.
- First, the momentum equation (4.38) reduces to Euler's equation:

$$\rho \frac{d\mathbf{V}}{dt} = \rho \mathbf{g} - \nabla p$$

Second, there is a great simplification in the acceleration term.

$$\frac{d\mathbf{V}}{dt} = \frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V} \cdot \nabla) \mathbf{V} \quad (\mathbf{V} \cdot \nabla) \mathbf{V} \equiv \nabla \left(\frac{1}{2} V^2 \right) + \boldsymbol{\zeta} \times \mathbf{V}$$

$$\left[\frac{\partial \mathbf{V}}{\partial t} + \nabla \left(\frac{1}{2} V^2 \right) + \boldsymbol{\zeta} \times \mathbf{V} + \frac{1}{\rho} \nabla p - \mathbf{g} \right] \cdot d\mathbf{r} = 0$$



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Nothing works right unless we can get rid of the third term.
We want

$$(\zeta \times \mathbf{V}) \cdot (d\mathbf{r}) \equiv 0$$

This will be true under various conditions:

1. $\mathbf{V}$ is zero; trivial, no flow (hydrostatics).
2. ζ is zero; irrotational flow.
3. $d\mathbf{r}$ is perpendicular to $\zeta \times \mathbf{V}$; this is rather specialized and rare.
4. $d\mathbf{r}$ is parallel to $\mathbf{V}$; we integrate along a streamline

If we integrate along a streamline in frictionless compressible flow and take,



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$$\frac{\partial \mathbf{V}}{\partial t} \cdot d\mathbf{r} + d\left(\frac{1}{2} V^2\right) + \frac{dp}{\rho} + g dz = 0$$

Except for the first term, these are exact differentials. Integrate between any two points 1 and 2 along the streamline:

$$\int_1^2 \frac{\partial V}{\partial t} ds + \int_1^2 \frac{dp}{\rho} + \frac{1}{2} (V_2^2 - V_1^2) + g(z_2 - z_1) = 0$$

This is Bernoulli's equation for frictionless unsteady flow along a streamline



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For incompressible steady flow, it reduces to

$$\frac{p}{\rho} + \frac{1}{2}V^2 + gz = \text{constant along streamline} \quad (4.120)$$

The constant may vary from streamline to streamline unless the flow is also irrotational.

For irrotational flow, the offending term vanishes regardless of the direction of dr , then holds all over the flow field with the same constant.



Velocity Potential

Irrotationality gives rise to a scalar function ψ similar and complementary to the stream function.

From a theorem in vector analysis, a vector with zero curl must be the gradient of a scalar function

$$\text{If } \nabla \times \mathbf{V} \equiv 0 \quad \text{then} \quad \mathbf{V} = \nabla \phi$$

where $\phi = \phi(x, y, z, t)$ is called the velocity potential function.

$$u = \frac{\partial \phi}{\partial x} \quad v = \frac{\partial \phi}{\partial y} \quad w = \frac{\partial \phi}{\partial z} \quad (4.122)$$

Note that ϕ , unlike the stream function, is fully three-dimensional and not limited to two coordinates.



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If a flow is both irrotational and described by only two coordinates, ϕ and ψ both exist and the streamlines and potential lines are everywhere mutually perpendicular except at a stagnation point.

A line of constant ϕ would be such that the change in ϕ is zero:

$$d\phi = \frac{\partial\phi}{\partial x} dx + \frac{\partial\phi}{\partial y} dy = 0 = u dx + v dy \quad (4.127)$$

Solving, we have

$$\left(\frac{dy}{dx}\right)_{\phi=\text{const}} = -\frac{u}{v} = -\frac{1}{(dy/dx)_{\psi=\text{const}}} \quad (4.128)$$



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Generation of Rotationality; A fluid flow that is initially irrotational may become rotational if

1. There are significant viscous forces induced by jets, wakes, or solid boundaries.

In this case Bernoulli's equation will not be valid in such viscous regions

2. There are entropy gradients caused by curved shock waves.
3. There are density gradients caused by stratification (uneven heating) rather than by pressure gradients.
4. There are significant noninertial effects such as the earth's rotation (the Coriolis acceleration).



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Internal flows, such as pipes and ducts, are mostly viscous, and the wall layers grow to meet in the core of the duct.

External flows, such as a body immersed in a stream, are partly viscous and partly inviscid, the two regions being patched together at the edge of the shear layer or boundary layer.

If the stream Mach number is less than about 0.3, we can combine Eq. (4.122) with the incompressible continuity equation (4.73):

$$\nabla \cdot \mathbf{V} = \nabla \cdot (\nabla \phi) = 0$$

$$\nabla^2 \phi = 0 = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} \quad (4.129)$$



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- **Example:** If a stream function and velocity potential exists for the velocity field $u=a(x^2-y^2)$, $v=-2axy$ and $w=0$. find it, plot it, and interpret it.
- **Solution:**
- **Example:** Consider the two-dimensional incompressible velocity potential $\phi=xy+x^2-y^2$, If it exists, find the stream function $\psi(x, y)$ of this flow



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- Solution step 1:** Note that this velocity distribution was also examined in Example 4.3. It satisfies continuity, Eq. (4.83), but let's check that; otherwise ψ will not exist:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = \frac{\partial}{\partial x}[a(x^2 - y^2)] + \frac{\partial}{\partial y}(-2axy) = 2ax + (-2ax) = 0 \quad \text{checks}$$

Thus we are certain that a stream function exists.

- Solution step 2:** To find ψ , write out Eqs. (4.85) and integrate:

$$u = \frac{\partial \psi}{\partial y} = ax^2 - ay^2 \quad (1)$$

$$v = -\frac{\partial \psi}{\partial x} = -2axy \quad (2)$$

and work from either one toward the other. Integrate (1) partially

$$\psi = ax^2y - \frac{ay^3}{3} + f(x) \quad (3)$$

Differentiate (3) with respect to x and compare with (2)

$$\frac{\partial \psi}{\partial x} = 2axy + f'(x) = 2axy \quad (4)$$

Therefore $f'(x) = 0$, or $f = \text{constant}$. The complete stream function is thus found:

$$\psi = a\left(x^2y - \frac{y^3}{3}\right) + C \quad \text{Ans. (5)}$$

To plot this, set $C = 0$ for convenience and plot the function

$$3x^2y - y^3 = \frac{3\psi}{a} \quad (6)$$

Since $w = 0$, the curl of $\mathbf{V}$ has only one z component, and we must show that it is zero:

$$\begin{aligned} (\nabla \times \mathbf{V})_z = 2\omega_z &= \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = \frac{\partial}{\partial x}(-2axy) - \frac{\partial}{\partial y}(ax^2 - ay^2) \\ &= -2ay + 2ay = 0 \quad \text{checks} \end{aligned} \quad \text{Ans.}$$

The flow is indeed irrotational. A velocity potential exists.

To find $\phi(x, y)$, set

$$u = \frac{\partial \phi}{\partial x} = ax^2 - ay^2 \quad (1)$$

$$v = \frac{\partial \phi}{\partial y} = -2axy \quad (2)$$

Integrate (1)

$$\phi = \frac{ax^3}{3} - axy^2 + f(y) \quad (3)$$

Differentiate (3) and compare with (2)

$$\frac{\partial \phi}{\partial y} = -2axy + f'(y) = -2axy \quad (4)$$

Therefore $f' = 0$, or $f = \text{constant}$. The velocity potential is

$$\phi = \frac{ax^3}{3} - axy^2 + C \quad \text{Ans.}$$



Thank you

end of chapter four